A Two-System Framework for Truth-Grade Three-Body Simulation and Sparse Discovery of Interpretable Predictive Dynamics

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Abstract

We present a two-system framework for studying and predicting three-body dynamics under coupled gravitational and electromagnetic interactions. The first system is a high-fidelity numerical simulator ("oracle") designed for reproducibility, performance, and explicit error control. It supports Newtonian gravity and a nonrelativistic electromagnetic interaction model including electric and magnetic fields as well as scalar and vector potentials (ϕ, A). The second system is a model-discovery engine that searches for sparse, interpretable equations that predict the oracle's trajectories with minimal complexity. In the current implementation, discovery proceeds via structured feature libraries and sparse regression (STLS/LASSO)[1, 5], using simulator-provided accelerations to avoid derivative noise; an optional GA baseline is also available. We provide an implementation-centric description and emphasize editorially the areas of modeling uncertainty: the electromagnetic approximation (quasi-static, no retardation or radiation reaction)[15], conservation-law interpretation under the chosen EM model, close-encounter numerical stiffness, and the limits of global predictability in chaotic regimes[6]. We argue that novelty arises not from claiming a universal closed-form solution, but from automatically discovering compact regime-specific laws with explicit validity domains and verified rollout performance against the oracle. Code repository: `https://github.com/Hendler/threebody`.

1 Introduction

The classical three-body problem is a paradigmatic nonlinear dynamical system exhibiting sensitivity to initial conditions and, for most configurations, the absence of closed-form global solutions. Practical prediction therefore relies on numerical integration, while theory offers partial analytic structure (integrals of motion, special solutions, perturbation expansions, invariant manifolds). For modern applications—ranging from celestial mechanics to charged particle dynamics—one often wants both: (i) an accurate simulator suitable as "ground truth," and (ii) interpretable approximations that predict well and illuminate structure.

We propose a two-system approach:

- **Truth-grade simulation system ("oracle")** – a careful, reproducible, numerically controlled engine that produces trajectories and physically meaningful diagnostics.
- **Equation discovery system ("predictor")** – a sparse identification pipeline that produces compact, interpretable predictive models that approximate the oracle and are validated by forward rollout.

This paper describes the architecture, numerical methods, and discovery methodology, and it explicitly lists modeling assumptions and unresolved issues.

2 Simulator ("Oracle") Design

2.1 State, coordinate choices, and invariants

The simulator evolves three bodies indexed by $i \in \{1, 2, 3\}$ with:

- Mass m_i, charge q_i.
- Position $r_i(t) \in \mathbb{R}^3$.
- Velocity $v_i(t) = \dot{r}_i(t) \in \mathbb{R}^3$.

The current Rust implementation is fixed to three bodies ($N = 3$) across simulation, output schema, and discovery; general N-body support is future work.

We typically re-center initial conditions to the center-of-mass (COM) frame and remove net linear momentum (barycentric initialization) to reduce drift. Invariants used as diagnostics include:

- Total linear momentum $P = \sum_i m_i v_i$.
- Total angular momentum $L = \sum_i r_i \times (m_i v_i)$.

A "mechanical energy" proxy (Section 2.4) is logged, with editorial caveats for EM.

2.2 Interaction model: gravity and nonrelativistic electromagnetism

Gravity (Newtonian). We use

$$a_i^{(g)} = \sum_{j \neq i} G m_j \frac{r_j - r_i}{\|r_j - r_i\|^3}.$$

Electromagnetism (chosen approximation). We include an explicit, nonrelativistic EM model.

Electric field at particle i:

$$E_i = \sum_{j \neq i} k_e q_j \frac{r_i - r_j}{\|r_i - r_j\|^3}.$$

Magnetic field at particle i via a quasi-static Biot–Savart-like term for moving charges:

$$B_i = \sum_{j \neq i} \frac{\mu_0}{4\pi} q_j \frac{v_j \times (r_i - r_j)}{\|r_i - r_j\|^3}.$$

Lorentz acceleration:

$$a_i^{(em)} = \frac{q_i}{m_i} \left(E_i + v_i \times B_i \right).$$

Total acceleration:

$$a_i = a_i^{(g)} + a_i^{(em)}.$$

We also compute potentials.

Scalar potential:

$$\phi_i = \sum_{j \neq i} k_e \frac{q_j}{\|r_i - r_j\|}.$$

Vector potential:

$$A_i = \sum_{j \neq i} \frac{\mu_0}{4\pi} \frac{q_j v_j}{\|r_i - r_j\|}.$$

Editorial uncertainty: This EM model omits finite propagation speed (retardation), radiation reaction, and self-field effects. It is best interpreted as a quasi-static approximation[15]; its domain of physical validity is not universal. For "truth-grade" claims, the simulator is truth-grade relative to the assumed equations, not necessarily relative to full Maxwell–Lorentz dynamics.

2.3 Numerical integration

We provide two integrators:

- **Adaptive RK45 (Dormand–Prince)**[7]: used as "truth mode" for controlled local truncation error. It adapts time step using embedded error estimates with user-specified tolerances and min/max dt.
- **Leapfrog (Kick–Drift–Kick)**[8, 9, 11, 10]: a fast, stable second-order method often favored for long-horizon gravitational dynamics.

We implement a strict "truth mode" that enables adaptive RK45 with step rejection and per-step dt tracking. If the embedded error exceeds tolerance, the step is rejected and dt is reduced; otherwise, dt is increased within configured bounds. This provides a best-effort accuracy strategy under the stated model assumptions, especially near close encounters where fixed-step methods can fail.

Editorial uncertainty: For purely Hamiltonian gravity, symplectic methods have strong long-term stability guarantees. With the chosen EM terms, the system is not treated with a full Hamiltonian field formulation here; leapfrog remains useful but should not be assumed to preserve the same structure exactly. This is one reason we maintain RK-based truth mode.

2.4 Diagnostics and conservation interpretation

We log $P(t)$ and $L(t)$, plus a "mechanical energy" diagnostic:

- Kinetic: $\sum_i \frac{1}{2} m_i \|v_i\|^2$.
- Gravitational potential: $\sum_{i<j} -G m_i m_j / \|r_{ij}\|$.
- Electric potential: $\sum_{i<j} k_e q_i q_j / \|r_{ij}\|$.

where $r_{ij} = r_j - r_i$.

Editorial uncertainty: In full electrodynamics, energy conservation includes field energy and radiation; in quasi-static approximations, interpreting "total energy" is subtle. We therefore treat energy drift as a numerical health metric, not a strict physical invariant under all EM configurations.

2.5 Close encounters and stiffness

Near-collisions can cause:

- Accelerations scaling as $1/r^2$ (and worse in some derived terms).
- Adaptive dt collapse.

- Large floating-point conditioning issues.

Current implementation supports optional softening ϵ in denominators to avoid singularities. In addition, when a close-encounter policy triggers softening, we can (optionally) ramp ϵ smoothly as a function of the minimum pair distance (to avoid discontinuous force jumps), and we can deterministically substep fixed-step integrators near close approaches (to reduce stiffness without changing the global output cadence). Encounter events record the minimum pair distance, the action taken, and (when applicable) the softening change and substeps used as part of the sidecar metadata.

Editorial uncertainty: These are pragmatic numerical stabilizers and still change the effective physics. A truly "truth-grade" approach for close encounters would incorporate regularization (e.g., KS/chain regularization)[16, 17] and/or higher precision. This remains a clear future-work item.

2.6 Output contract for downstream discovery

The simulator outputs a wide CSV including:

- r_i, v_i for each body.
- a_i (total acceleration) for each body.
- E_i, B_i, ϕ_i, A_i.
- Diagnostics: energy proxy, P, L, drifts.

Design rationale: Discovery should use simulator-provided accelerations, not finite differences of noisy trajectories.

3 Discovery Engine ("Predictor"): Sparse, Interpretable Dynamics

3.1 Objective

We seek simple, interpretable functions $\hat{f}_i$ that approximate oracle accelerations, building on work in automated equation discovery and sparse model identification[3, 4, 1]:

$$a_i(t) \approx \hat{f}_i\left(\text{features}(r(t), v(t), \ldots)\right),$$

with high forward-prediction fidelity over a time horizon, minimal equation complexity, and explicit domain-of-validity.

This is fundamentally different from "solving the three-body problem in closed form." The goal is regime-specific compact predictive structure. Related work extends sparse discovery to spatiotemporal systems and PDE settings.[2]

3.2 Vector targets and component-wise discovery

In implementation, we fit acceleration components separately (e.g., $a_{i,x}$, $a_{i,y}$, $a_{i,z}$) using a shared feature library, then combine the component equations into a vector-valued model. This preserves interpretability while enabling full trajectory rollout evaluation.

3.3 Why we fit acceleration directly

Classical system identification often estimates $\dot{x}$ from measurements; in chaotic systems this introduces strong noise amplification. We avoid this by training on simulator-provided $a_i(t)$ rather than differentiating $v_i(t)$.

3.4 Feature libraries (Θ) as structured hypothesis spaces

We build vector-valued basis functions ("terms") from physically meaningful atoms:

- Relative vectors $r_{ij} = r_j - r_i$.
- Distances $d_{ij} = \|r_{ij}\|$.
- Unit directions $\hat{r}_{ij}$.
- Velocities and differences v_i, v_{ij}.
- Cross products capturing magnetic-like structure $v_j \times r_{ij}$.
- Triple products capturing Lorentz-like structure $v_i \times (v_j \times r_{ij})$.

We implement libraries of increasing expressivity:

- **default** (current implementation): direction-preserving, component-wise bases that mirror the analytic force contributions *without* physical constants:
  - Gravity basis: $g_i^* = \sum_{j \neq i} m_j (r_j - r_i)/\|r_j - r_i\|^3$.
  - Electric basis: $e_i^* = (q_i/m_i) \sum_{j \neq i} q_j (r_i - r_j)/\|r_i - r_j\|^3$.
  - Magnetic basis: $b_i^* = (q_i/m_i)\, v_i \times B_i^*$, where $B_i^* = (1/4\pi) \sum_{j \neq i} q_j (v_j \times (r_i - r_j))/\|r_i - r_j\|^3$.

  Under our quasi-static model, this yields $a_i \approx G\, g_i^* + k_e\, e_i^* + \mu_0\, b_i^*$ (with softening applied consistently).

- **extended** (current implementation): adds a small number of carefully chosen atoms to expand expressivity while preserving interpretability, including (i) additional distance scalings (e.g., $1/r^4$ variants), and (ii) simple distance-based gating ("close" vs. "far" features) to help isolate regime-specific structure without resorting to an unconstrained polynomial soup.

- **em_fields** (current implementation): a minimal EM library that learns Lorentz acceleration directly from simulator fields. Features are in acceleration units (already include physical constants), so in EM-enabled runs we expect coefficients near 1:
  - Electric term: $\ell_{E,i} = (q_i/m_i) E_i$.
  - Magnetic term: $\ell_{v \times B,i} = (q_i/m_i)(v_i \times B_i)$.

  This supports a simple discovered form $a_i \approx G\, g_i^* + 1 \cdot \ell_{E,i} + 1 \cdot \ell_{v \times B,i}$ (with consistent softening).

Editorial note: The library design is where most "new math" leverage lives. If the library is too broad, discovery becomes unstable and non-interpretable; too narrow, it cannot capture dynamics. We currently err toward small, interpretable libraries.

3.5 Sparse discovery via sparse regression (current implementation)

Let y be a target acceleration component (e.g., $a_{i,x}$) and let Θ be the feature matrix evaluated over time (columns are library terms evaluated at each time step). We restrict to models linear in features,

$$y \approx \Theta\xi,$$

and in the current Rust implementation we estimate ξ on a fixed library via sparse regression (retaining the top-3 equations per component across a small hyperparameter path):

- **STLS** (sequential thresholded least squares): normalize library columns, solve ridge/OLS for ξ, hard-threshold small coefficients below τ, refit on the surviving support, and iterate until the support stabilizes.
- **LASSO**: coordinate descent along an ℓ_1 path parameterized by α (warm-started), yielding sparse ξ on a fixed library.
- A lightweight genetic algorithm (GA) over sparse coefficient vectors remains available as an optional baseline; it is slower and non-deterministic but provides an alternative search dynamic.

In addition to per-component coefficient estimation, the CLI's `factory` loop maintains a small *population* of full vector equations across iterations: it carries forward the incumbent best and a few recent elites, injects a handful of axis-consistent mutations (an equation-level GA), and re-scores them under new initial conditions. This makes candidate equations the primary object of iteration while keeping numeric rollout metrics as the selection criterion.

Related work: Sparse regression (SINDy/STLS)[1] and LASSO-style solvers[5] are natural coefficient estimators for fixed libraries and are implemented in the Rust codebase; the data contract (header-based CSV plus sidecar metadata) supports reproducible regression and search-based baselines.

3.6 What counts as "predicts well"

Pointwise acceleration fit is not sufficient. We evaluate:

- Rollout accuracy: integrate the discovered model forward and compare trajectory error (e.g., position RMSE) against the oracle.
- Stability and failure modes (blow-ups, divergence time).
- Generalization to held-out initial conditions within the intended regime.

Editorial uncertainty: In chaotic regimes, long-horizon point prediction is not meaningful[6]; evaluation should shift toward short-horizon deterministic accuracy, ensemble/statistical predictions, and qualitative invariants and manifold structure. In the current implementation we partially address this with predictability tooling: (i) small ensembles of nearby initial conditions to measure conditioning and detect when the ensemble agrees on an emergent binary/outcome (*"lock points"*), and (ii) event-driven extraction of close-encounter minima plus a lightweight sparse baseline that models energy and angular-momentum exchange across encounters. These tools aim to quantify *when* predictions become well-conditioned, without claiming universal long-horizon microstate predictability.

3.7 LLM-assisted interpretation (optional)

We add an optional LLM judge layer to interpret candidate equations and propose next experiments. The LLM does *not* override numeric ranking; it is a supplemental reviewer that produces a structured scorecard and narrative synthesis. Concretely, it decides (a) proposed initial conditions within bounds, (b) an interpretable ranking of candidates under a fixed rubric, and (c) next-step recommendations (rollout integrator, discovery solver/hyperparameters, and search directions), while the discovery engine performs coefficient estimation/search (STLS/LASSO or an optional GA baseline). The rubric is explicit and machine-readable, with component scores for (i) predictive fidelity, (ii) parsimony, (iii) physical plausibility, (iv) regime consistency, and (v) stability risk. Evidence passed to the judge includes equation forms plus numeric metrics (MSE, rollout RMSE, divergence time) and regime diagnostics.

Optionally, the judge may also propose a concrete candidate equation written directly in the project's feature language (a vector form for $a_{i,x}$, $a_{i,y}$, $a_{i,z}$ over the currently selected feature library) and/or recommend switching between small feature libraries (`default`, `extended`, `em_fields`). These proposals are treated as hypotheses: they are parsed, validated, and scored by the same numeric evaluators before they are considered "better". We require JSON-only output to preserve determinism, and all prompts and responses are logged as reproducibility artifacts. To stabilize iterative progress, we additionally maintain a global "best so far" ledger over local runs (e.g., `results/best_results.md` and `results/best_results.json`), grouped by comparable simulation buckets (steps, dt). The factory loop can include the incumbent best equation and its metrics in the LLM context as a concrete target to beat, while keeping numeric selection as the source of truth. The judge can additionally recommend (a) which rollout integrator to use for evaluation (Euler vs. leapfrog), (b) which fitness heuristic to use for candidate selection (MSE vs. MSE with a parsimony penalty), and (c) which discovery solver/hyperparameters to try next (STLS/LASSO/GA), enabling controlled ablations without compromising reproducibility.

4 System Integration and Reproducibility

4.1 CLI-driven workflow

The systems are controlled by CLI:

- **simulate**: produce truth trajectories and diagnostics.
- **discover**: fit sparse equations using a chosen library.
- **predictability**: ensemble runs, lock-point detection, close-encounter extraction, and a sparse encounter-map baseline (experimental).
- **factory**: an optional LLM-guided loop that selects initial conditions, runs simulation, performs discovery, and emits a judged report with suggested next runs.
- **quickstart**: a convenience wrapper that runs an end-to-end workflow and writes a novice-friendly summary plus evidence under a single directory.
- **bench-em**: deterministic EM identifiability suite (charge/velocity sweeps + diverse initial conditions) that writes a comparable report per case.
- **findings**: render an aggregated LaTeX report of best-discovered equations across local runs (from the best-results ledger).

- **predict** (planned): roll out discovered equations and compare to truth.
- **score** (planned): compute quantitative benchmarks.
- **render-gif** (script): render a simple GIF animation from a trajectory CSV (header-based parsing).

4.2 LLM-guided "factory" loop (optional)

The factory loop operationalizes a scientific workflow: (1) propose initial conditions, (2) simulate, (3) discover equations, (4) judge results under a fixed rubric, and (5) decide whether to continue. The LLM assists only in steps (1) and (4), and all artifacts (config, initial conditions, prompts, responses, metrics) are saved per iteration to ensure reproducibility and auditability. At the end of a run, we additionally render reproducible evaluation artifacts (Markdown and LaTeX) that record the best discovered equations (exact text), the best-run initial conditions, and a best-vs-history comparison over prior local attempts. We also save machine-readable numeric evidence: per-iteration rollout error curves (`rollout_trace.json`) and a fixed held-out benchmark report (`benchmark_eval.json`). The global best-results ledger is also updated to capture the current incumbent per simulation bucket.

4.3 Robust I/O design

We explicitly fixed a key fragility: rigid CSV schemas break as output evolves. We therefore parse by header name and extract only needed columns for any given task. This enables rapid iteration.

4.4 Determinism and metadata

A "truth-grade" oracle requires:

- Fixed integrator and tolerance settings recorded.
- Initial condition specifications recorded.
- Constants and force toggles recorded.

Current implementation outputs CSV and a JSON "sidecar" containing config, header hash, warnings, last-regime diagnostics, dt statistics (accepted/rejected/min/max/avg), close-encounter event metadata (when triggered), and the initial state (masses/charges/positions/velocities) to support strict reproducibility.

5 Results (What we expect, what we can claim, and what we cannot)

5.1 Expected outcomes

For gravity-only or weak-EM regimes:

- **gravity_min** should rediscover near-inverse-square directional structure.
- **em_basic** may recover Lorentz-like cross terms if EM effects are present and sufficiently sampled.
- Sparse models can achieve strong short-horizon rollout fidelity with few terms.

5.2 What we can claim

- The simulator provides consistent numerical solutions to the implemented equations with user-controlled accuracy (RK45 mode).
- The discovery engine can identify compact, interpretable approximations within structured libraries and validate them by rollout.

5.3 What we cannot honestly claim (yet)

- That the simulator is "physically exact" for EM in all regimes (it is quasi-static).
- That energy conservation is strict in combined EM dynamics as implemented (field energy not modeled).
- That close-encounter accuracy is truth-grade without regularization/high precision.
- That a single global simple formula predicts chaotic three-body dynamics long-horizon.

6 Limitations and Open Questions (Editorial)

This section is intentionally frank.

Electromagnetic modeling fidelity

The quasi-static magnetic approximation may be adequate for some regimes but fails when retardation or radiation are significant. If the objective is "physics truth," the EM model must be upgraded (retarded potentials or field solver), at significant computational cost.

Integrator structure under EM

Leapfrog is attractive for gravity; under EM it may not preserve the right structure. We currently treat leapfrog as a fast option, not as a guaranteed long-term invariant-preserving method under full coupled dynamics. EM-appropriate schemes include the Boris pusher and structure-preserving charged-particle integrators.[12, 13, 14]

Close encounters and stiffness

Softening (even when smoothed) is not a principled solution. We also support deterministic substepping near close encounters, but a truly truth-grade approach likely requires regularization (Sundman transform, KS/chain regularization) and possibly higher precision.

Discovery stability and identifiability

Sparse regression can select misleading terms when features are correlated or data is not sufficiently rich. Better experimental design (diverse initial conditions, active learning) and more robust optimization (e.g., constrained solvers) are likely necessary. In practice, a common failure mode is to repeatedly rediscover gravity-only structure even when EM is enabled. We therefore log a simple identifiability diagnostic on each oracle trajectory: the ratio $\mathbb{E}(|a_{\mathrm{em}}|)/\mathbb{E}(|a_{\mathrm{grav}}|)$. If this ratio is $\ll 1$, EM terms are unlikely to be selected by sparse regression under the chosen regimes and noise levels. To operationalize "rich enough" data, we include a deterministic EM benchmark suite (`bench-em`)

that sweeps charge scale and velocity scale across multiple initial-condition templates to produce EM-heavy regimes where EM terms should become identifiable.

Meaning of "novel math"

We interpret "novel" as compact approximations that outperform naive baselines within a regime and are expressed in human-interpretable terms. We do not claim a universal analytic solution.

7 Future Work

- Add principled close-encounter regularization (KS/chain, Sundman transforms) and/or multi-precision truth mode (beyond softening/smooth substepping).
- Implement predict and score commands with standardized benchmarks (divergence time, RMS error curves).
- Implement regime partitioning ("atlas of models"): train multiple sparse models with a lightweight regime classifier.
- Extend discovery to probabilistic forecasts for chaotic regimes.
- Extend the LLM judge rubric and experiment-planning prompts (ablation suggestions, regime partitioning), while keeping numeric scoring primary.

8 Conclusion

We described a pragmatic two-system approach to three-body dynamics: a high-fidelity simulator as oracle, and a sparse discovery engine that produces interpretable, compact predictive equations. The central technical strategy is to train on simulator-provided accelerations and to constrain the hypothesis space through physically structured feature libraries, thereby producing "simple math" that predicts well within explicit regimes. We also highlighted the primary uncertainties: electromagnetic approximation fidelity, conservation-law interpretation, close-encounter stiffness, and the limits of long-horizon prediction in chaotic systems. This framework is designed to be extensible, reproducible, and honest about what is known versus assumed.

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A Implementation Notes (brief)

- CSV parsing must be header-based to remain stable as columns evolve.
- Discovery expands vector bases into x/y/z component features and estimates sparse linear combinations via STLS/LASSO sparse regression (with an optional GA baseline).